

OCOM5100M

Programming for Data Science

Data Analysis Project Topics

[Contents](#)

OCOM5100M 1

Programming for Data Science..... 1

 Data Analysis Project Topics 1

 1. Air Accidents Analysis (2008-2022)..... 2

 2. UK Online Retail E-commerce Data 2

 3. S&P 500 Stock Returns (2009-2019) 2

 4. England’s Unpaid Care Workforce 3

 5. Obesity and Lifestyle Factors 3

 6. UFO Sighting Reports in the United States 3

 7. Video Game Player Retention Factors 3

 8. Diabetes Health Indicators 4

 9. Formula 1 Championship Data (1950 – 2020)..... 4

 10. Global Happiness and Wellbeing..... 4

 11. Retail Shopping Behaviour and Fraud Detection..... 4

 12. Urban Air Quality Analysis (1958 – 2025)..... 5

 13. UK Earthquakes (1995 – 2024) 5

 14. Flight Delays and Cancellations (2024)..... 5

- 15. Global Cybersecurity Threats (2015 – 2024)..... 5
- 16. Breast Cancer Gene Expression and Clinical Data 6
- 17. Global Climate Change Indicators 6
- 18. Netflix Movies and TV Shows 6
- 19. Spotify Music Characteristics and Popularity 6
- 20. World University Rankings 7
- 21. Global Energy Consumption and Production..... 7

1. Air Accidents Analysis (2008-2022)

Explore temporal trends in accident frequency and fatalities across different time periods. Analyse patterns by flight phase, aircraft characteristics, and engine types. Investigate geographical distribution of accidents and compare against global air traffic densities. Examine correlations between aircraft attributes and safety outcomes through statistical analysis.

Dataset: [Air Accidents from Jan2008 to May2022](#)

2. UK Online Retail E-commerce Data

Identify high-performing products and key sales markets to understand demand patterns. Analyse revenue trends across different timeframes to detect seasonal variations and peak periods. Segment customers based on purchasing behaviours including frequency, value, and product preferences. Develop insights for inventory optimization and marketing strategies.

Dataset: [E-commerce data](#)

3. S&P 500 Stock Returns (2009-2019)

Visualize key financial metrics across S&P 500 companies segmented by industry and sector. Evaluate relationships between financial ratios and stock price movements. Apply predictive modelling techniques to forecast stock performance using historical data. Identify potential leading indicators of company performance.

Dataset: [S&P 500 Stock Returns](#)

4. England's Unpaid Care Workforce

Analyse regional distribution of unpaid carers across England. Investigate gender disparities in unpaid care provision. Examine age group contributions to the unpaid care workforce. Explore relationships between deprivation levels and care provision patterns.

Dataset: [Office of National Statistics](#)

5. Obesity and Lifestyle Factors

Conduct exploratory analysis to identify lifestyle attributes correlating with weight levels. Apply regression techniques to examine relationships between numerical and categorical variables and BMI. Determine the most relevant features influencing obesity through unsupervised learning methods. Validate causality between behaviours and weight outcomes.

Dataset: [Estimation of Obesity Levels based on eating habits and physical condition](#)

6. UFO Sighting Reports in the United States

Analyse trends in sighting frequency and geographical distribution across the US. Examine temporal patterns and seasonal variations in reported sightings. Investigate characteristics of first-hand accounts and witness descriptions. Explore potential correlations with weather events and urban density.

Dataset: [Kaggle: UFO Sightings](#)

Secondary Datasets:

- [Kaggle: United States Cities Database](#)
- [Kaggle: US Weather Events Data](#)

7. Video Game Player Retention Factors

Compare player engagement between single-player and multiplayer games. Examine the relationship between achievement systems and average playtime. Identify common

characteristics among top-performing games with highest player retention. Analyse tags and features associated with successful games.

Dataset: [Kaggle: Steam Games Dataset](#)

8. Diabetes Health Indicators

Identify indicators most strongly associated with diabetes diagnosis. Investigate how demographic factors influence predictive variables. Apply machine learning models to predict diabetes development. Examine interactions between multiple health indicators.

Dataset: [Kaggle: Diabetes Health Indicators Dataset](#)

9. Formula 1 Championship Data (1950 – 2020)

Create comprehensive visualizations of race results and driver performance. Develop integrated dashboards combining multiple performance metrics. Analyse historical trends in racing outcomes. Compare performance across different eras and rule changes.

Dataset: [Kaggle: Formula 1 World Championship 1950 - 2020](#)

10. Global Happiness and Wellbeing

Visualize and compare happiness levels across countries and regions. Quantify relative contributions of various factors to national wellbeing including GDP, social support, health, freedom, generosity, and corruption. Analyse chronological trends in happiness from 2011-2024. Investigate correlations between different happiness factors. Compare regional trajectories and identify socio-economic drivers of wellbeing changes.

Dataset: [World Happiness Report](#)

11. Retail Shopping Behaviour and Fraud Detection

Analyse customer spending patterns and demographic influences on purchasing behaviour. Detect fraud patterns using risk metrics across different payment methods. Evaluate relationships between consumer behaviour and fraud likelihood. Identify satisfaction indicators and their connections to transaction characteristics.

Datasets: [Kaggle: Shopping behaviour dataset](#)

12. Urban Air Quality Analysis (1958 – 2025)

Classify and visualize temporal trends in $PM_{2.5}$ and NO_2 concentrations across cities. Examine long-term changes and seasonal patterns in pollution levels. Identify cities with highest average pollution and those showing significant improvements. Assess correlations between different pollutants across locations and time periods.

Dataset: [Kaggle: Urban Air Quality and Climate Dataset](#)

13. UK Earthquakes (1995 – 2024)

Analyse changes in recorded earthquake frequency over three decades. Examine magnitude distribution and identify most severe seismic events. Investigate geographical distribution patterns across the UK. Explore temporal clustering and seasonal variations if present.

Dataset: [Earthquakes Recent UK Events Data](#)

14. Flight Delays and Cancellations (2024)

Analyse patterns and trends in flight cancellations by time period. Identify causes and duration patterns of flight delays. Determine which airports experience the most operational disruptions. Examine relationships between delay causes and severity.

Datasets: [Kaggle: Flight delay and cancellation data](#)

15. Global Cybersecurity Threats (2015 – 2024)

Analyse financial impact across different attack types and threat categories. Measure effectiveness of various defence mechanisms and response times. Identify most targeted industries by country and region. Examine temporal trends in threat evolution and mitigation success rates.

Dataset: [Kaggle: Global Cybersecurity Threats \(2015 – 2024\)](#)

16. Breast Cancer Gene Expression and Clinical Data

Integrate RNA-sequencing gene expression data with clinical information for key genes (BRCA1, BRCA2, TP53, ERBB2, ESR1). Investigate differential expression patterns between tumour and normal samples. Assess subtype-specific variations across molecular classifications. Evaluate correlations between gene expression and clinical parameters including patient age and tumour stage. Generate visualizations to illustrate expression patterns and clinical associations.

Dataset: [Xena: Breast Cancer Gene Expression](#)

17. Global Climate Change Indicators

Analyse global temperature changes across different regions and time periods. Examine trends in climate indicators such as CO₂ emissions, sea level rise, and extreme weather events. Compare climate change impacts between developed and developing nations. Investigate correlations between industrial activity and environmental changes. Assess seasonal and annual variations in temperature anomalies.

Dataset: [Kaggle: Temperature change](#)

18. Netflix Movies and TV Shows

Explore content distribution across genres, countries, and release years. Analyse trends in content production and platform additions over time. Investigate differences between movie and TV show characteristics including duration and ratings. Examine geographical patterns in content creation. Identify popular genres and their evolution across different time periods.

Dataset: [Kaggle: Netflix Shows](#)

19. Spotify Music Characteristics and Popularity

Analyse audio features that correlate with track popularity including tempo, energy, and valence. Explore genre distributions and their characteristic sound profiles. Investigate temporal trends in music characteristics across decades. Examine relationships between artist

popularity and track features. Compare audio patterns across different music genres and cultural regions.

Dataset: [Kaggle: Ultimate Spotify Tracks Database](#)

20. World University Rankings

Compare university performance metrics across different ranking systems. Analyse geographical distribution of top-performing institutions. Investigate correlations between various performance indicators such as research output, teaching quality, and international outlook. Examine trends in institutional rankings over time. Explore relationships between university characteristics and overall scores.

Dataset: [Kaggle: World University Rankings](#)

21. Global Energy Consumption and Production

Analyse energy consumption patterns across countries and regions over time. Compare renewable versus non-renewable energy source adoption rates. Investigate relationships between economic development and energy usage. Examine trends in energy efficiency and carbon intensity. Explore disparities in per capita energy consumption across different income levels and geographical areas.

Dataset: [Kaggle: World Energy Consumption](#)